

Preclass Learning: Loops and Iteration P1 (ICA 11)

1. The for Loop

A for loop is used to execute a block of code a fixed number of times, typically by iterating over the elements of a sequence (vector, list, or range).

- **Syntax:** `for (item in sequence) { code_to_execute_for_each_item }`
- **Use Cases:**
 - Applying a transformation to every file in a directory.
 - Iterating through a list of models or parameters.
 - Performing a simulation a set number of times.

2. The while Loop

A while loop is used to execute a block of code repeatedly **as long as a specified logical condition remains TRUE**.

- **Syntax:** `while (condition_is_TRUE) { code_to_execute; update_condition }`
- **Use Cases:**
 - Repeating a process until convergence (e.g., optimization algorithms).
 - Running an interactive game or prompt until the user enters a specific command.
- **Caution:** If the condition is never updated or never becomes FALSE, the loop will run indefinitely (infinite loop).

3. Alternatives to Loops

While loops are fundamental, R code often favors **Vectorization** (performing operations on entire vectors at once) and **Functional Programming** (using map functions) because they are often faster and result in cleaner code.